

Storage system for optimizing LLM checkpointing in HPC

Advanced Computing Project 25/26

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Context

LLM training spans several weeks, requiring periodic checkpointing to a Parallel File System (PFS) to prevent total state loss during node failures.

However, checkpointing directly to the PFS creates two problems:

- Simultaneous writes create heavy I/O contention, causing the "noisy neighbor" problem for other jobs.
- High write latency stalls training, leading to significant compute time being wasted.

Problem

Challenge	Impact
PFS Contention	I/O bottlenecks slow training as nodes compete for bandwidth.
Resilience	Hardware failures at scale risk losing weeks of training progress.
Checkpointing	Larger models → larger checkpoints → higher storage costs/time.
Storage Costs	Frequent PFS writes increase costs and operational overhead.

Table 1: Challenges of LLM Checkpointing to Parallel File Systems

Motivation

The goal is to ... to achieve:

- **TODO:** TODO

Approach

- TODO

Architecture

- TODO

Results

- TODO

Next Steps

- TODO